

CBSE 2027 Physics — Predicted Paper (Set 5 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. A cylinder of radius R and length L is placed in a uniform electric field $\vec{E}$ parallel to the cylinder's axis. The total electric flux through the entire surface of the cylinder is: (a) $2\pi R^2 E$ (b) $\pi R^2 E$ (c) Zero (d) $2\pi RLE$ (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q2. The storage battery of a car has an electromotive force (EMF) of 12 V . If the internal resistance of the battery is $0.4\ \Omega$, what is the maximum current that can be drawn from the battery? (a) 15 A (b) 30 A (c) 12 A (d) 20 A (Chapter: Current Electricity) | **[Confidence: Very High]**

Q3. A straight conducting rod of length l and mass m is suspended in a horizontal plane by a pair of flexible strings in a uniform magnetic field of magnitude B . To remove the tension in the supporting strings, the magnitude of the current I that must be passed through the rod is: (a) $\frac{mg}{lB}$ (b) $\frac{mgl}{B}$ (c) $\frac{lB}{mg}$ (d) $\frac{mgB}{l}$ (Chapter: Moving Charges and Magnetism) | **[Confidence: High]**

Q4. An electromagnet is a current-carrying solenoid with a core. Based on their B-H curve properties, the most suitable magnetic material for making the core of an electromagnet is one having: (a) High retentivity and high coercivity (b) Low retentivity and low coercivity (c) High permeability, low retentivity, and low coercivity (d) Low permeability, high retentivity, and high coercivity (Chapter: Magnetism and Matter) | **[Confidence: Very High]**

Q5. The magnetic flux linked with a coil (in Weber) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of the induced EMF in the coil at time $t = 4\text{ s}$ will be: (a) 27 V (b) 43 V (c) 108 V (d) 210 V (Chapter: Electromagnetic Induction) | **[Confidence: High]**

Q6. In a series LCR circuit, the voltage across the inductor, capacitor, and resistor are 20 V , 20 V , and 40 V respectively. The voltage of the applied AC source is: (a) 40 V (b) 80 V (c) $40\sqrt{2}\text{ V}$ (d) 20 V (Chapter: Alternating Current) | **[Confidence: Very High]**

Q7. Which of the following electromagnetic waves is primarily used to treat muscular strain? (a) X-rays (b) Ultraviolet rays (c) Infrared radiation (d) Microwaves (Chapter: Electromagnetic Waves) | **[Confidence: Medium]**

Q8. An object is placed at 10 cm in front of a concave mirror of radius of curvature 15 cm . The position of the image is: (a) -30 cm (b) $+30\text{ cm}$ (c) -15 cm (d) $+15\text{ cm}$ (Chapter: Ray Optics and Optical Instruments) | **[Confidence: High]**

Q9. In a Young's double-slit experiment, the path difference at a certain point on the screen between two interfering waves is $\frac{1}{8}^{th}$ of the wavelength ($\lambda/8$). The ratio of intensity at this point to the maximum intensity at the centre of a bright fringe is: (a) 0.853 (b) 0.707 (c) 0.500 (d) 0.942 (Chapter: Wave Optics) | **[Confidence: High]**

Q10. A photocell connected in an electrical circuit is placed at a distance d from a light source, resulting in a photocurrent I . What will be the photocurrent when the distance is reduced to $d/2$? (a) I (b) $2I$ (c) $4I$ (d) $I/4$ (Chapter: Dual Nature of Radiation and Matter) | **[Confidence: Very High]**

Q11. The energy of an electron in the n^{th} orbit of a hydrogen atom is given by $E_n = -\frac{13.6}{n^2} \text{ eV}$. The negative sign of the energy indicates that: (a) the electron is free to move. (b) the electron is tightly bound to the nucleus. (c) the kinetic energy of the electron is equal to its potential energy. (d) the atom is continuously radiating energy. (Chapter: Atoms) | **[Confidence: Very High]**

Q12. In an extrinsic semiconductor, the number density of holes is $4 \times 10^{20} \text{ m}^{-3}$. If the number density of intrinsic carriers is $1.2 \times 10^{15} \text{ m}^{-3}$, the number density of electrons in it is: (a) $3.6 \times 10^9 \text{ m}^{-3}$ (b) $2.4 \times 10^{10} \text{ m}^{-3}$ (c) $1.8 \times 10^9 \text{ m}^{-3}$ (d) $3.2 \times 10^{10} \text{ m}^{-3}$ (Chapter: Semiconductor Electronics) | **[Confidence: High]**

Q13. Assertion (A): If the distance between the parallel plates of a capacitor is halved, its capacitance is doubled. **Reason (R):** The capacitance of a parallel plate capacitor is inversely proportional to the separation between its plates. (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q14. Assertion (A): A flexible conducting wire bent into an irregular shape tends to enclose a maximum circular area when a strong current is passed through it. **Reason (R):** Opposite currents flowing in adjacent parts of the wire repel each other, pushing the wire outwards into a circular shape. (Chapter: Moving Charges and Magnetism) | **[Confidence: Medium]**

Q15. Assertion (A): In Young's double-slit experiment, all interference fringes formed on the screen are of strictly equal width. **Reason (R):** The fringe width (β) depends on the wavelength of light (λ), the distance of the screen from the slits (D), and the slit separation (d). (Chapter: Wave Optics) | **[Confidence: Very High]**

Q16. Assertion (A): The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energies (from zero up to a maximum). **Reason (R):** The energy of electrons emitted from deeper inside the metal surface is partially lost in collisions with other atoms before they can escape. (Chapter: Dual Nature of Radiation & Matter) | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. A steady current of 8 mA flows through a metallic wire. Calculate the exact number of electrons passing through any cross-section of the wire in **10 seconds**. (Chapter: Current Electricity) | **[Confidence: Very High]**

Q18. A straight horizontal wire of mass 200 g and length 1.5 m carries a current of 2 A . It is suspended motionless in mid-air by a uniform horizontal magnetic field B perpendicular to

the wire. Calculate the magnitude of the magnetic field. (Take $g = 9.8 \text{ m/s}^2$). (Chapter: Moving Charges and Magnetism) | **[Confidence: High]**

Q19. Two identical circular loops, P and Q , each of radius 5 cm , are lying in mutually perpendicular planes with a common centre. Find the magnitude and direction of the net magnetic field at their common centre if they carry currents of 3 A and 4 A respectively. (Chapter: Moving Charges and Magnetism) | **[Confidence: High]**

Q20. What is the ratio of the velocity of two light waves of wavelengths $4000 \text{ \AA}$ and $8000 \text{ \AA}$ while travelling in a vacuum? Give a very brief justification for your answer. (Chapter: Electromagnetic Waves / Ray Optics) | **[Confidence: Very High]**

Q21. A fast-moving neutron collides with the nucleus of Plutonium (${}^{239}_{94}\text{Pu}$), undergoing nuclear fission to produce Xenon (${}^{134}_{54}\text{Xe}$) and Zirconium (${}^{103}_{40}\text{Zr}$) along with additional neutrons. Write the complete, balanced nuclear reaction for this process. (Chapter: Nuclei) | **[Confidence: Medium]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. A parallel plate capacitor of capacitance C is fully charged by a battery to a potential difference V . The battery is then completely disconnected. Now, the distance between the plates is doubled, and a dielectric slab of dielectric constant $K = 1.5$ is inserted to fill the entire space between the plates. How will the following be affected? (i) The electric field between the plates. (ii) The total electrostatic energy stored in the capacitor. Provide mathematical justification for both. (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q23. Two primary cells of EMF E_1 and E_2 having internal resistances r_1 and r_2 respectively, are connected in parallel, with their positive terminals connected together. Derive an expression for the equivalent EMF and equivalent internal resistance of the combination. (Chapter: Current Electricity) | **[Confidence: Very High]**

Q24. State Bohr's quantization condition for angular momentum. Using this, obtain an expression for the magnetic dipole moment of an electron revolving in the n^{th} circular orbit of a hydrogen atom. (Chapter: Atoms / Magnetism and Matter) | **[Confidence: Very High]**

Q25. Derive an expression for the total impedance of a series LCR circuit connected to an AC voltage source $V = V_0 \sin(\omega t)$. Draw the corresponding phasor diagram showing the voltage vectors. (Chapter: Alternating Current) | **[Confidence: Very High]**

Q26. Plot a graph showing the variation of the angle of deviation (δ) as a function of the angle of incidence (i) for a ray of light passing through a glass prism. Hence, derive the formula for the refractive index (μ) of the material of the prism. (Chapter: Ray Optics and Optical Instruments) | **[Confidence: Very High]**

Q27. State three important features of Einstein's photoelectric equation. Plot a graph showing the variation of the maximum kinetic energy (K_{max}) of emitted photoelectrons against the frequency (ν) of incident radiation. How can we determine Planck's constant from this graph? (Chapter: Dual Nature of Radiation and Matter) | **[Confidence: Very High]**

Q28. Draw a typical plot showing the potential energy of a pair of nucleons as a function of their separation distance r . Write two important characteristic conclusions regarding nuclear forces that can be drawn from this graph. (Chapter: Nuclei) | **[Confidence: High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: Dielectrics in Capacitors *Passage:* When a dielectric slab is placed in an external uniform electric field, the molecules of the dielectric align themselves, resulting in the polarization of the dielectric. This creates an internal induced electric field opposite to the external field, effectively reducing the net electric field inside the material. This property is heavily utilized in capacitors to increase their capacity to store charge. (a) Define the dielectric constant (K) of a medium in terms of the electric field. (1 mark) (b) What happens to the capacitance of a parallel plate capacitor if a dielectric slab of thickness equal to the plate separation (d) is introduced between the plates? (1 mark) (c) A parallel plate air capacitor has a capacitance of **10 pF** and is connected to a **50 V** battery. While the battery remains connected, the space between the plates is filled with a dielectric ($K = 4$). Calculate the new charge on the plates. **OR** If the battery was disconnected *before* inserting the same dielectric, calculate the new potential difference across the plates and the new energy stored. (2 marks) (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q30. Case Study 2: The Electromagnetic Spectrum *Passage:* The electromagnetic spectrum encompasses a continuous range of frequencies and wavelengths of electromagnetic waves. James Clerk Maxwell's theoretical prediction of displacement current led to the realization that light is an electromagnetic wave. Different parts of the spectrum, from radio waves to gamma rays, interact with matter differently and have unique methods of production and varied practical applications. (a) Name the part of the electromagnetic spectrum which is most suitable for radar systems used in aircraft navigation. (1 mark) (b) Identify the electromagnetic waves that are strongly absorbed by the ozone layer in the stratosphere, protecting life on Earth. (1 mark) (c) How are X-rays typically produced? State one major medical or industrial use of X-rays. **OR** Why are infrared waves often referred to as "heat waves"? Explain their role in maintaining the Earth's average temperature via the greenhouse effect. (2 marks) (Chapter: Electromagnetic Waves) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Define electric flux. State Gauss's Law in electrostatics. **(b)** A spherical conducting shell of inner radius R_1 and outer radius R_2 has a net charge Q . A point charge q is placed exactly at the centre of the shell. Determine the surface charge density on the (i) inner surface, and (ii) outer surface of the shell. **(c)** Write the expression for the electric field at a point located at a distance $r > R_2$ from the centre. **OR (a)** Derive an expression for the potential energy of a system of two point charges q_1 and q_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively in a uniform external electric field $\vec{E}$. **(b)** Three point charges $+q$, $-4q$, and $+2q$ are placed at the three vertices of an equilateral triangle of side ' a '. Calculate the exact amount of work done required to dissociate the system and move these charges to infinite distances apart. (Chapter: Electric Charges and Fields / Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q32. (a) State the underlying principle of a moving coil galvanometer. **(b)** What is a radial magnetic field? Explain how it is produced in a galvanometer and state its specific significance. **(c)** A galvanometer with a coil resistance of $120\ \Omega$ shows full-scale deflection for a current of $2.5\ \text{mA}$. Calculate the value of the resistance required to convert it into an ammeter of range 0 to $7.5\ \text{A}$. **OR (a)** A metallic rod of length l is rotated with an angular velocity ω in a uniform magnetic field B acting perpendicular to the plane of rotation. One end of the rod is hinged at the centre, and the other sweeps the circumference. Derive the expression for the EMF induced across the ends of the rod. **(b)** A circular coil of N turns and cross-sectional area A is rotated at a constant angular speed ω in a uniform horizontal magnetic field B . Obtain the expression for the maximum EMF induced in the coil. (Chapter: Moving Charges and Magnetism / Electromagnetic Induction) | **[Confidence: Very High]**

Q33. (a) Define a wavefront. Use Huygens' principle to draw the refracted wavefront and verify Snell's law of refraction for a plane wavefront propagating from a rarer medium into a denser medium. **(b)** In a Young's double-slit experiment, the separation between the slits is $0.5\ \text{mm}$ and the screen is placed $1.5\ \text{m}$ away. If the wavelength of the monochromatic light used is $600\ \text{nm}$, calculate the exact distance between the central bright fringe and the 4th dark fringe on the screen. **OR (a)** Draw a clear, labelled ray diagram of a refracting Astronomical Telescope in normal adjustment (when the final image is formed at infinity). **(b)** Derive the expression for the magnifying power of the telescope in normal adjustment. **(c)** A small telescope has an objective lens of focal length $140\ \text{cm}$ and an eyepiece of focal length $5.0\ \text{cm}$. Calculate the magnifying power of the telescope for viewing distant stars. What is the length of the telescope tube in this adjustment? (Chapter: Wave Optics / Ray Optics and Optical Instruments) | **[Confidence: Very High]**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.